

Customer Churn Prediction Using Temporal Behavioural Features

178,440	8,506	0.754	0.868
Rows	Customers	Holdout F1	Holdout Recall

Executive summary

This case study builds a customer churn prediction system for FoodCorp using time-aware behavioural features. Churn is defined as no further purchase within 35 days of a weekly reference date. The modelling design intentionally separates past behaviour used for features from future behaviour used only for labels, so the analysis reflects an operational weekly scoring workflow rather than a retrospective customer description.

The final modelling table contains 178,440 customer-week observations across 8,506 customers and 75 weekly reference dates. The last 8 weeks are held back as an untouched test period; model selection and tuning are performed only on earlier temporal validation folds. A tuned Logistic Regression pipeline is selected over Random Forest because it provides stronger validation F1 and recall, while remaining interpretable enough to explain behavioural drivers of churn.

On the final holdout period, the cleaned Logistic Regression pipeline achieves accuracy of 0.782, precision of 0.667, recall of 0.868 and F1 score of 0.754. The strongest practical insight is that churn is driven primarily by weakening shopping regularity and narrowing category breadth, not by spend alone.

Project snapshot

Business question	Which active customers are at elevated risk of becoming inactive, and what behaviours explain that risk?
Analytical approach	SQL feature engineering, weekly temporal labelling, manual out-of-time validation, tuned Logistic Regression.
Target definition	Churn = no purchase within the next 35 days after an active weekly reference date.
Model objective	Prioritise early identification of at-risk customers while keeping the model interpretable for marketing and technical teams.

1. Churn definition and threshold selection

A churn label must be commercially meaningful before any predictive model is useful. A threshold that is too short flags normal shopping gaps as churn; a threshold that is too long identifies disengagement too late for intervention. Four beta thresholds were compared to balance responsiveness with practical grocery shopping variation.

Beta (days)	Avg active customers	Avg churn events	Avg churn % among active
21	1,004.76	42.87	4.29%
28	1,155.11	36.47	3.20%
35	1,281.62	32.26	2.56%
42	1,392.65	29.24	2.14%

The 35-day threshold is the selected operating definition: it is less aggressive than 28 days but more responsive than 42 days. This definition is then used consistently for labelling, feature engineering and model evaluation.

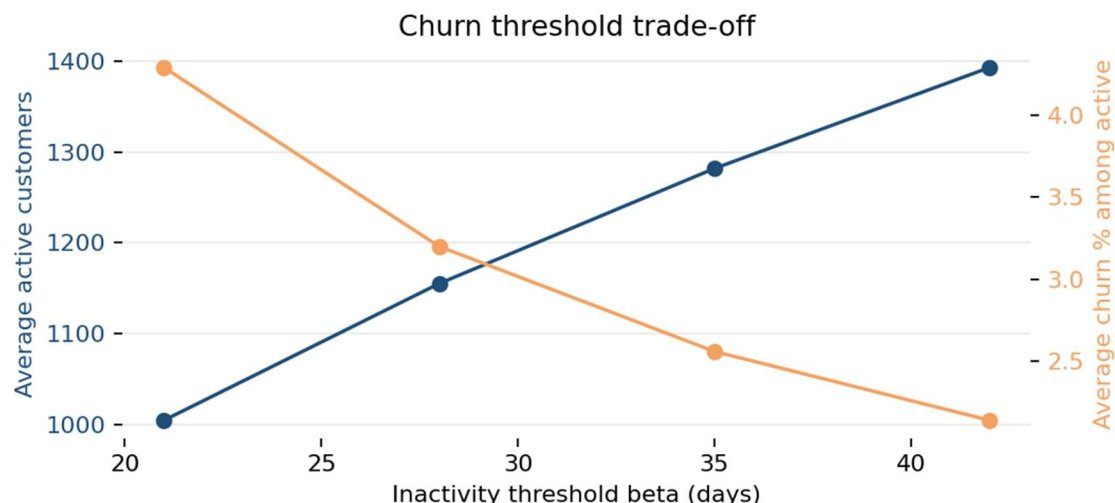


Figure 1. Trade-off between active base size and churn rate across beta thresholds.

2. Temporal feature engineering and leakage control

The modelling unit is one customer at one weekly reference date. For each reference date, features are calculated only from information available on or before that date, while the churn label uses future behaviour after that date. This design prevents future purchase information from leaking into the model inputs.

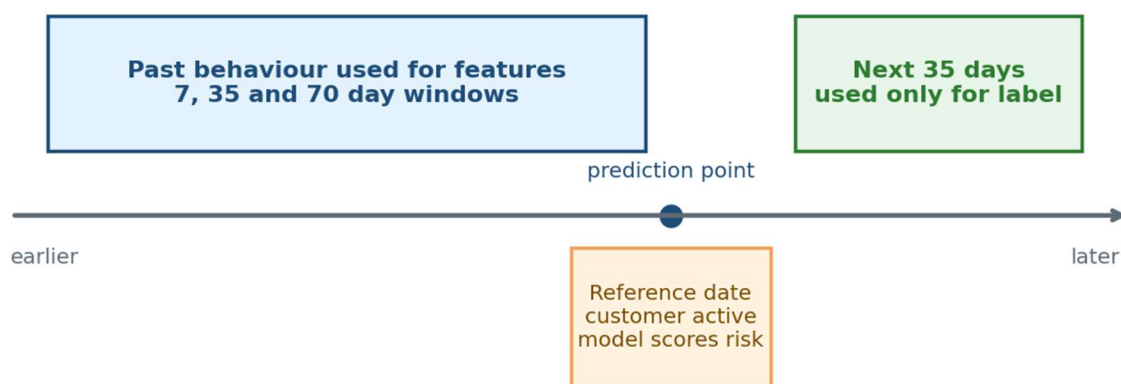


Figure 2. Past windows are used for features; future data is reserved only for label creation.

Feature block	Examples	Business meaning
Recency	recency_days	How close the customer is to the inactivity threshold.
Frequency	purchase_days_7d, purchase_days_35d, purchase_days_70d	Regularity of shopping over short and medium horizons.
Monetary value	spend_7d, spend_35d, spend_70d	Recent customer value and spending continuity.
Quantity / volume	qty_7d, qty_35d, qty_70d	Basket volume beyond cash spend alone.
Category breadth	category_count_35d, category_count_70d	How broadly the customer is engaged with FoodCorp categories.
Trend variables	purchase_days_change_35d, spend_change_70d	Whether behaviour is weakening or improving versus the prior comparable period.

Four exact duplicate variables were removed before final modelling: purchase_days_recent_35d, purchase_days_recent_70d, spend_recent_35d and spend_recent_70d. Row-definition fields such as customer_id, ref_date and beta_days were retained for dataset construction but excluded from model inputs.

3. Model development and evaluation

A random train-test split was avoided because the problem is temporal. Instead, the final 8 weeks were frozen as an untouched holdout period. Model comparison and tuning used three expanding temporal validation folds inside the development period.

Fold	Training window	Validation window
1	First 43 weeks	Next 8 weeks
2	First 51 weeks	Next 8 weeks
3	First 59 weeks	Final 8 weeks of development sample

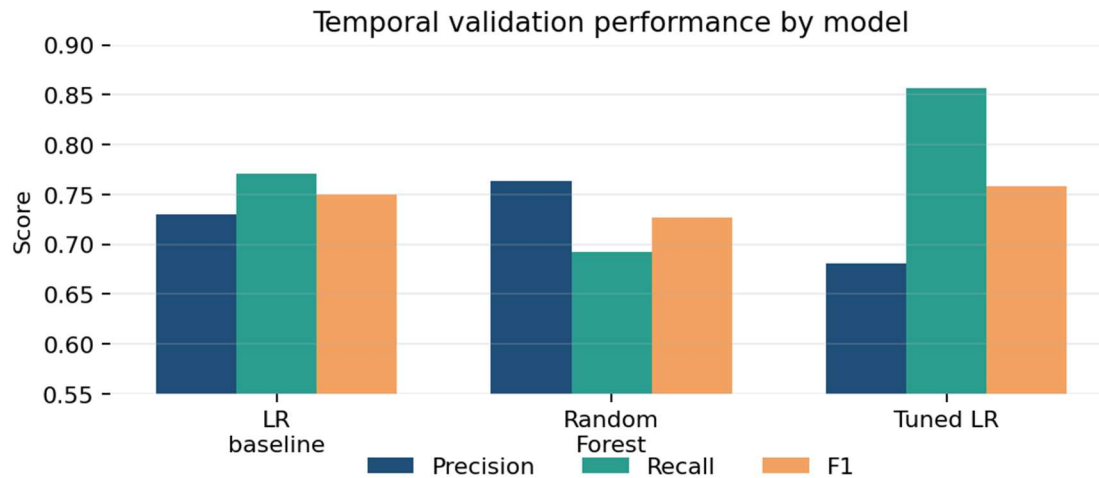


Figure 3. Logistic Regression achieved a stronger F1-recall balance than Random Forest in temporal validation.

Model	Accuracy	Precision	Recall	F1
Logistic Regression baseline	0.7949	0.7298	0.7711	0.7498
Random Forest	0.7921	0.7638	0.6928	0.7265
Tuned Logistic Regression	-	0.6808	0.8565	0.7586

The final model is a StandardScaler + LogisticRegression pipeline with $C = 3.0$, $\text{class_weight} = \text{"balanced"}$, $\text{max_iter} = 1000$ and a 0.50 decision threshold. The 0.50 threshold was retained after tuning because it achieved the strongest validation F1.

4. Final holdout performance

Accuracy	Precision	Recall	F1 score
0.7824	0.6667	0.8681	0.7542

Untouched holdout confusion matrix

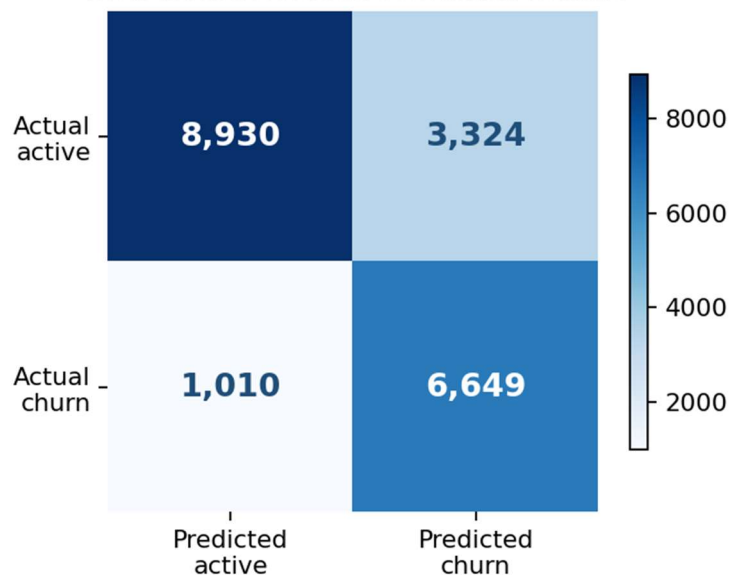


Figure 4. Holdout confusion matrix: 6,649 churners identified and 1,010 churners missed.

The model deliberately prioritises recall because missing a customer who is already moving towards inactivity is commercially costly. Precision of 0.667 means the model will still flag some customers who do not churn, so campaign economics and contact capacity should determine how aggressively FoodCorp acts on the scores.

5. Model interpretation and business insight

The interpretation uses two complementary views. Standardised Logistic Regression coefficients show direction: negative coefficients reduce churn risk, while positive coefficients increase churn risk. Permutation importance shows reliance: how much F1 drops when a feature is disrupted.

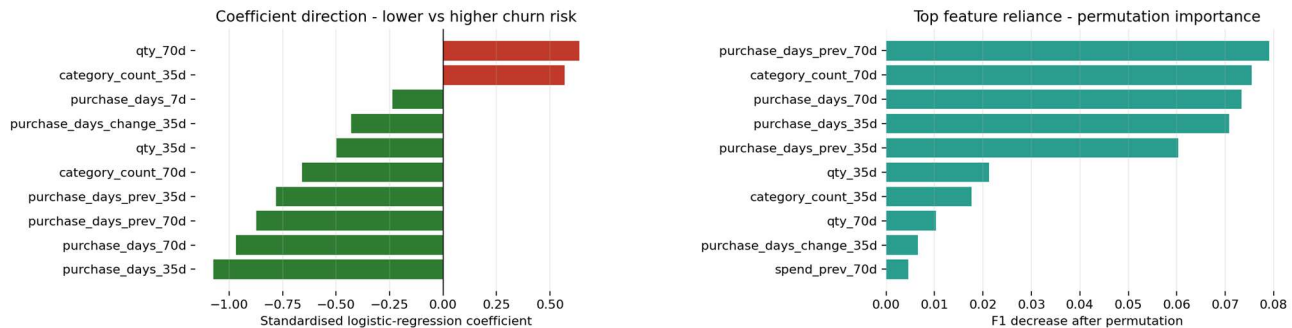


Figure 5. Coefficients explain direction; permutation importance explains model reliance.

The clearest pattern is behavioural regularity. Higher values for `purchase_days_35d`, `purchase_days_70d` and previous-period purchase-day variables reduce churn risk, while disrupted shopping frequency has the largest practical impact on model performance. Category breadth is also central: `category_count_70d` has a strong negative coefficient and high permutation importance, indicating that customers buying across more categories are more stable.

Spend and quantity contribute signal, but they are less important than frequency and category breadth. This is an important marketing message: a customer can still spend occasionally while their routine is already weakening. FoodCorp should monitor regularity and breadth, not only spend.

6. Deployment recommendations

Recommendation	Action	Rationale
Weekly risk scoring	Score active customers every week using only pre-reference-date behaviour.	Matches the temporal design and enables timely intervention before full inactivity.
Segment intervention tiers	Use probability bands such as high, medium and low risk rather than a single operational rule.	Allows campaign spend to match expected risk and customer value.
Regularity-focused offers	Trigger reminders or personalised offers when purchase-day frequency weakens.	The model shows regular shopping behaviour is the strongest loyalty signal.
Category-breadth protection	Encourage customers to maintain multi-category shopping missions.	Narrowing category engagement is a warning sign of relationship weakening.
Monitor model drift	Track precision, recall, F1 and calibration by month before retraining.	Shopping behaviour, seasonality and campaign strategies may change over time.

7. Limitations and next steps

- The output is predictive, not causal. A feature associated with churn does not prove that changing it will prevent churn.
- The current model uses behavioural transaction features only. Customer demographics, channel interactions, promotion exposure and customer-service signals could improve targeting if available.
- The model should be validated prospectively on future campaign periods and monitored for drift before full operational deployment.
- Marketing actions should be A/B tested to estimate incremental retention impact, not only response among high-risk customers.